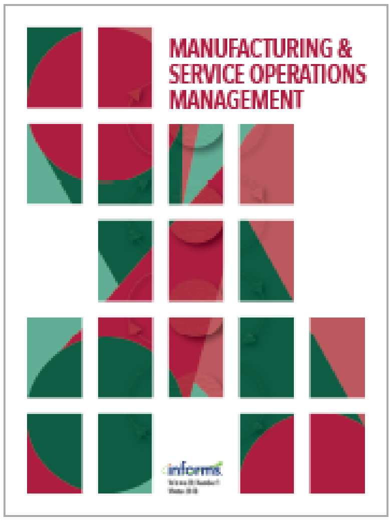

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The Value of Health Information Technology Interoperability: Evidence from Interhospital Transfer of Heart Attack Patients

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Abstract. ***Problem definition*:** Health information technology (HIT) *interoperability* refers to the ability of different electronic health record systems and software applications to communicate, exchange data, and use the information that has been exchanged. The U.S. Government has invested heavily to promote HIT interoperability in recent years in an attempt to improve patient outcomes and control healthcare expenditure. This study empirically assesses the value of HIT interoperability in the interhospital transfer process of heart attack patients. ***Academic/practical relevance*:** HIT interoperability is supposed to enable health information exchange between disparate providers. However, there exists little evidence about how it affects care delivery processes *across* providers. ***Methodology*:** Using transfer records of heart attack patients and HIT interoperability adoption records of hospitals in New York State between 2013 and 2015, we estimate the effect of HIT interoperability on care delivery process measures and patient outcome measures. We demonstrate the robustness of the results with alternative samples and model specifications, including the instrumental variable method, the propensity score matching method, the difference-in-differences method, and a falsification test. ***Results*:** We show that HIT interoperability shortens the throughput time of interhospital transfer by 45.6 minutes on average or 12.0%. Surprisingly, we find that HIT interoperability has little effect in reducing duplicate electrocardiogram (EKG) testing for transferred patients at receiving hospitals. When HIT interoperability is enabled through a common software vendor, it yields 15.6% more reduction in the throughput time than when it is enabled through different vendors, but it still has no significant effect on duplicate EKG testing. Furthermore, we find that HIT interoperability leads to a 3.0-percentage point decrease in the 30-day readmission rate of transferred patients, which can be explained by the reduction in the throughput time. ***Managerial implications*:** Our findings demonstrate the value of incentivizing HIT adoption and promoting widespread exchange of health information because HIT interoperability indeed improves the efficiency of healthcare delivery across providers, which ultimately translates to improved patient outcomes. Given the lack of reduction in duplicate testing in our study, we call for more provider effort toward realizing the full potential of HIT interoperability by minimizing the gap between technology adoption and utilization.

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Keywords: interoperability • health information exchange • heart attack • interhospital patient transfer • throughput time • duplicate testing

1. Introduction

The rise of health information technology (HIT), particularly electronic health record (EHR) systems, has made it possible for healthcare providers to get access to medical records at anytime from anywhere. Since the passing of the Health Information Technology for Economic and Clinical Health (HITECH) Act in 2009, EHR adoption in the U.S. healthcare industry has gained momentum. By 2015, 84% of hospitals had adopted at least a basic EHR system, which represents a ninefold increase since 2008 (Henry et al. 2016). Despite the rapid adoption of EHRs among U.S. healthcare providers, widespread exchange of health information across providers has not yet occurred because of the lack of interoperability between disparate providers. HIT interoperability refers to the ability of different EHR systems and software applications to communicate, exchange data, and use the information that has been exchanged (HIMSS 2013).

Achieving HIT interoperability has become increasingly important as today's healthcare services are delivered across a complex network of providers, including primary care clinics, community hospitals, tertiary hospitals, laboratories, and medical imaging facilities. Sharing health information across providers and reducing unnecessary repeat tests and procedures are two promised benefits of EHR adoption.[1] Without acquiring HIT interoperability, these benefits are difficult to realize as healthcare providers are handicapped by not being able to exchange health information with other providers electronically. In the last few years, the U.S. Government has heavily promoted HIT interoperability in an attempt to improve patient outcomes and control healthcare expenditure. In 2011, the Centers for Medicare and Medicaid Services established the Medicare and Medicaid EHR Incentive Programs (now known as the Promoting Interoperability Programs) to encourage providers to adopt certified EHR technologies and demonstrate meaningful use of HIT.[2] In 2015, U.S. Congress declared "a national objective to achieve widespread exchange of health information through interoperable certified EHR technology nationwide by December 31, 2018."[3] The current status of interoperability in the U.S. healthcare system is still far from achieving this ambitious goal. According to the Office of the National Coordinator for Health Information Technology, only 41% of hospitals nationwide routinely access health information from other providers, and less than half of hospitals integrate the data they receive from other providers into their own EHR systems.[4] In 2015, fewer than 30% of hospitals were engaged in all four domains of interoperability, which are finding, sending, receiving, and integrating clinical information (Holmgren et al. 2017).

Despite the importance of HIT interoperability in national healthcare policy making, there exists little evidence about how HIT interoperability might transform care delivery processes and whether such transformation will be translated into improvement of patient outcomes. The existing literature has primarily focused on the impact of HIT interoperability *within* individual healthcare providers and has found some evidence of improved operational efficiency and patient outcomes (e.g., Ayer et al. 2019, Chen et al. 2019). What is missing in the existing literature is the effect of HIT interoperability on care delivery processes *across* healthcare providers. Given that HIT interoperability is the capability to exchange health information between healthcare providers electronically, understanding how it impacts care delivery processes beyond the boundary of healthcare organizations will help realize the promised benefits of improving patient outcomes and reducing healthcare expenditure. In this study, we aim to fill this knowledge gap by delving into a critical care process for heart attack patients that spans across providers—interhospital patient transfer.

When a patient experiences heart attack, an ambulance will be dispatched and send the patient to the emergency department (ED) of a nearby hospital. It has been reported that 12.4% of all heart attack patients in the United States ended up being transferred from the EDs they initially encountered to different hospitals to undergo revascularization procedures (Lu and Lu 2018). Percutaneous coronary intervention (PCI) and coronary artery bypass grafting (CABG) are the two most commonly applied revascularization procedures, which mechanically open the clogged artery in the heart and reestablish vascular connection and blood perfusion. Many community hospitals in the United States lack the capability to conduct revascularization, thereby causing the high rate of transfer for heart attack patients.

We adopt the context of interhospital transfer of heart attack patients to study the effect of HIT interoperability for three reasons. First, interhospital transfer is a critical process in the care for heart attack patients. The American Heart Association recommends that the time lapse between severe heart attack patients' first medical contact and their revascularization treatment should be less than 120 minutes (Lu and Lu 2018). Therefore, timely transfer of heart attack patients to the right provider for revascularization treatment is essential to patient outcomes because "time is muscle." Second, interhospital transfer enables us to quantify the effect of HIT interoperability on a process that spans across hospitals. Third, because a heart attack patient is usually sent by an ambulance to the nearest hospital,[5] this setting naturally provides a quasirandom assignment of patients to different hospitals, thereby mitigating the issue of patient selection of hospitals.

Using hospital transfer records of heart attack patients and HIT interoperability adoption records of hospitals in the New York State between 2013 and 2015, we estimate the effect of HIT interoperability on two care delivery process measures: (1) the throughput time of interhospital transfer, which is an indicator of the operational efficiency of the transfer process, and (2) duplicate electrocardiogram (EKG) testing (EKG is an essential test to diagnose heart attack). Our results show that HIT interoperability speeds up the interhospital transfer process by shortening 45.6 minutes (or 12.0%) of the throughput time, which is the duration between a patient's admission to a sending hospital's ED and the patient's admission to a receiving hospital. Surprisingly, our results show that HIT interoperability has little effect on reducing duplicate

EKG testing for transferred patients at receiving hospitals. These results are robust to alternative samples and model specifications as well as different estimation methods, including the instrumental variable (IV) method, the propensity score matching (PSM) method, the difference-in-differences (DID) method, and a falsification test. Our heterogeneous analysis shows that when HIT interoperability is enabled through a common software vendor, it yields 15.6% more reduction in the throughput time than when it is enabled through different vendors, but it still has no significant effect on duplicate EKG testing. Furthermore, we find that HIT interoperability leads to a 3.0-percentage point decrease in the 30-day readmission rate of transferred patients, which can be explained by the reduction of the throughput time.

Our findings demonstrate the value of incentivizing HIT adoption and promoting widespread exchange of health information because HIT interoperability indeed improves the efficiency of healthcare delivery across providers, which ultimately translates to improved patient outcomes. To the best of our knowledge, this work is the first to demonstrate the value of HIT interoperability from the perspective of care delivery processes *across* disparate providers. Hence, our study helps to open the black box of how HIT interoperability improves healthcare outcomes. Given the lack of reduction in duplicate testing in our study, we call for more provider effort toward realizing the full potential of HIT interoperability by minimizing the gap between technology adoption and utilization.

2. Background and Hypothesis Development

2.1. Interhospital Transfer of Heart Attack Patients

Acute myocardial infarction (AMI), commonly known as heart attack, is the leading cause of death in the United States, accounting for one in four deaths currently. About every 40 seconds, someone in the United States experiences a heart attack.[6] Most heart attacks occur without prior warnings, and patients are typically sent to the ED of the nearest hospitals via emergency services. Unfortunately, many community hospitals in the United States lack the resources to conduct revascularization procedures such as PCI and CABG to treat heart attack. For example, PCI, a preferred nonsurgical procedure of revascularization for patients with a severe form of heart attack (e.g., ST-elevation myocardial infarction, medically known as STEMI), is only available in 36.2% of the hospitals in the United States (Concannon et al. 2012). Compared with clot-busting medication, revascularization has been widely recognized for its superiority on STEMI patients' health outcomes. Therefore, when an STEMI patient needs revascularization treatment but cannot obtain it in the hospital he initially encounters, the hospital would transfer him to a nearby hospital with revascularization capability immediately.

To overcome the problem of limited revascularization availability in community hospitals, coordinated interhospital transfer is advocated to improve the quality of care for STEMI patients (Rokos et al. 2006, Shah et al. 2008). For transferred STEMI patients, the American College of Cardiology Foundation and the American Heart Association recommend that the first medical contact to device time should be less than 120 minutes. However, achieving timely transfer has been facing obstacles. It is reported that only 10% of the STEMI patients undergoing interhospital transfer receive the treatment within the recommended time frame (Veinot et al. 2012), and much time delay is spent on arranging patient transfer (Aguirre et al. 2008). In addition, inefficiencies in interhospital transfer processes, such as duplicate testing because of poor communication between hospitals, high load of photocopying medical records, and limited transportation resource availability, have been reported (Gupta et al. 2010, Bosk et al. 2011, Feazel et al. 2015). These observations raise the question of whether acquiring HIT interoperability might help to improve the efficiency of interhospital transfer.

2.2. Interoperability and Health Information Exchange in the New York State

Digitalization of health information holds the promise of transforming the way healthcare is delivered. In 2009, the U.S. Congress passed the HITECH Act as part of the American Recovery and Reinvestment Act, which earmarked $19.2 billion as incentives for healthcare providers to adopt EHRs. EHR adoption has proliferated in recent years. In 2008, the basic EHR adoption rate was at or below 22% in all states, whereas by 2015, the adoption rate increased above 65% across all states, and in 35 states, the adoption rate was above 80% (Henry et al. 2016).

With huge amounts of digital health records being created and stored in fragmented healthcare networks, electronic sharing of health information across providers is facing challenges because of the lack of interoperability between different EHR systems. Because EHR systems are provided by various software vendors, different data format and technology infrastructure may hinder providers from exchanging health information directly through their existing EHR systems. To enable interoperability, health information exchange (HIE) software solutions have been developed.

HIE is the acronym for health information exchange, and it refers to the *software application* that enables the electronic health information exchange functionality for a provider. HIE can also be used to refer to a *network* of providers that have the capability to exchange health information electronically. There are two major types of HIE networks (Arzt 2018). First, a *community* HIE achieves interoperability at a region or state level by adopting software applications that link disparate EHR systems within the community. Second, an EHR-*vendor-hub* HIE enables interoperability between two providers that adopt interoperable EHR systems from the same vendor. For example, the largest EHR company in the United States, Epic, provides the functionality of HIE to its customers that adopt its EHR system.

The Department of Health of New York established a state-wide community HIE network in 2008, which is known as the Statewide Health Information Network for New York (SHIN-NY). Between 2008 and 2015, SHIN-NY was composed of nine regional health information organization, each of which has adopted HIE applications to allow providers to access patient records in other providers' EHR systems within each region. Healthcare providers in the state have the option to participate in the network of SHIN-NY.[7] Figure A1 in the online appendix presents an example of the web portal of Healthix, a regional HIE network in the state of New York. Through the web portal, doctors can easily get access to patients' previous medical records, such as medication records and laboratory testing results, even if they are not stored in their own EHR systems. By 2015, 79.8% of hospitals in the state of New York had acquired HIE applications offered by various vendors, according to our data.

2.3. Hypothesis Development

As explained in Section 2.1, many community hospitals in the United States lack the resources to conduct revascularization procedures to treat heart attack and thus, need to transfer patients to revascularization-capable hospitals. According to the interviews conducted by Veinot et al. (2012) at three community hospitals, the process to transfer heart attack patients typically consists of five key steps: (1) *diagnosis*: conduct an EKG test on a patient to diagnose heart attack; (2) *decision*: decide whether the patient should be transferred;[8] (3) *negotiation*: contact candidate hospitals and negotiate a transfer; (4) *transportation*: choose a transportation method and contact either ambulance or helicopter services; and (5) *handoff*: prepare the EKG chart and all relevant medical records and transport and hand off the patient to the receiving hospital.

We conjecture that HIT interoperability might improve the efficiency of the transfer process in the following two aspects. First, HIT interoperability facilitates the negotiation step by improving the communication efficiency between sending and candidate receiving hospitals. Inadequate communication and inaccurate information exchange between hospitals are common in interhospital transfers (Feazel et al. 2015). Although the transfer process is typically protocolized, the negotiation between a sending hospital and a receiving hospital can be frustrating and time consuming, and the key challenge of the negotiation process is the communication inefficiency between hospitals (Bosk et al. 2011). After contacted by a sending hospital about transferring a patient, a potential receiving hospital will identify a physician to accept the patient. Relevant clinical records, such as current laboratory tests, radiology images, medical history, progress notes, or discharge summary, are needed for the physician at the receiving hospital to discuss with the physician at the sending hospital about whether it is appropriate to transfer the patient. In the absence of HIT interoperability, sharing clinical records requires the medical staff in the sending hospital to talk over the phone or send emails to the receiving hospital. The inefficiency of phone calls has been a challenge during the transfer process. For example, one emergency physician described "her role in caring for patients in need of transfer as often waiting by the phone to ensure that all the information was given to make the transfer possible—rather than being at the bedside of the patient" (Bosk et al. 2011). Such communication inefficiency could be alleviated by HIT interoperability because the staff members at receiving hospitals can retrieve needed patient information through their HIE software applications in a seamless and accurate manner, thereby leading to a shortened transfer negotiation step.

Second, HIT interoperability facilitates the handoff step by automating the transfer of clinical records and speeding up the handoff communication. According to Bosk et al. (2011), physically transmitting all needed clinical records between a sending hospital and a receiving hospital is a rather difficult task. They conduct a microlevel analysis of interhospital transfers by interviews. They find that after a receiving hospital agrees to the transfer of a patient, the sending hospital needs to prepare the patient's clinical records to be transferred with him to the receiving hospital, which involves laborious photocopying or scanning of documents in the absence of HIT interoperability. Heart attack patients have many relevant clinical records that need to be transferred, which include the time of acute symptom onset, the duration of pain, history of prior AMI/stent/CABG/renal failure, and EKG testing results, etc.[9] With HIT interoperability, the time for the sending hospital to prepare and send clinical records and the time for the receiving hospital to retrieve them can be significantly reduced because they can be shared digitally through HIE software applications. If indeed HIT interoperability improves the efficiency of the negotiation and handoff steps in the transfer process, we conjecture that it would shorten the throughput time of interhospital transfer, as stated in the following hypothesis.

Hypothesis 1. *Heart attack patients transferred between interoperable hospitals experience a shorter throughput time of the interhospital transfer process than patients transferred between noninteroperable hospitals.*

As anecdotal evidence indicates, transferred patients' documentation and diagnostic test results may not be readily available to doctors at receiving hospitals, thereby leading to duplicate testing (Nacht et al. 2013). EKG is a quick, painless test that provides valuable diagnosis of the heart's electrical activity. When a patient arrives at the ED with heart attack symptoms, the ED physician will order the EKG test and rely on the EKG result to make treatment and transfer decisions for the patient. According our data, 93.9% of all transferred heart attack patients received EKG tests in sending hospitals. In addition, doctors can distinguish STEMI patients from non-STEMI patients based on their EKG results because there is an apparent ST-segment elevation in STEMI patients' EKGs. According to standard cardiac care protocols, STEMI patients will definitely be transferred to revascularization-capable hospitals for immediate treatment (Veinot et al. 2012).

If the receiving hospital has an accurate copy of the EKG test, then it is possible to not repeat EKG testing because the last one has been done within a few hours. Stewart et al. (2010) find that duplicate testing for transferred patients is common, and interoperable EHR systems would help to reduce duplicate testing. If two hospitals are interoperable, staff at receiving hospitals will be able to retrieve patient records electronically through their HIE software applications, which would, in turn, reduce duplicate testing. Indeed, prior studies have shown that adoption of HIE is associated with reduction of imaging service utilization, such as computed tomography scan (CT scan), x-ray, and ultrasound (Lammers et al. 2014, Ayabakan et al. 2017). However, as far as we know, no studies have investigated the effect of interoperability on repeat EKG tests for *transferred* patients at receiving hospitals. Thus, we formulate the following hypothesis.

Hypothesis 2. *Heart attack patients transferred between interoperable hospitals are less likely to receive duplicate EKG testing in receiving hospitals than patients transferred between noninteroperable hospitals.*

Finally, timely transfer of heart attack patients to revascularization-capable hospitals for revascularization treatment is essential to patient outcomes because "time is muscle." After a patient experiences a heart attack, it is vital to remove the blockage in the heart vessel as soon as possible. As every minute goes by, more heart muscle is going to die. Therefore, reducing the throughput time of the transfer process will allow heart attack patients to receive revascularization treatment sooner, thereby improving their health outcomes, as found by many medical studies (Chan et al. 2012, Estévez-Loureiro et al. 2013, Farshid et al. 2015). Moreover, the electronic sharing of medical information itself can lead to improved health outcomes (Ayer et al. 2019, Chen et al. 2019). Based on these findings in the literature, we formulate the following hypothesis.

Hypothesis 3. *Heart attack patients transferred between interoperable hospitals have better health outcomes than patients transferred between noninteroperable hospitals.*

3. Literature Review

Our work contributes to two streams of literature in healthcare management: (1) the use of technology in healthcare delivery and (2) patient routing.

3.1. The Use of Information Technology in Healthcare Delivery

Our work directly contributes to the literature on HIE and HIT interoperability. It is worth noting that in most contexts, HIE and HIT interoperability are interchangeable concepts, but different authors might prefer one term over the other. As HIE is becoming a trend in practice, there is a growing stream of literature studying its impact on healthcare cost and quality. Walker et al. (2005) examine the value of HIE by establishing a cost-benefit model based on published evidence with expert opinions. Their estimation results show that fully implemented HIE yields a net value of $77.8 billion in the United States annually. Frisse et al. (2011) estimate that HIE access is associated with a total annual cost savings of $1.9 million for all regional hospital EDs in Memphis, Tennessee. Magnus et al. (2012) show that public HIE is associated with improved health outcomes for HIV patients. Vest et al. (2014) find that access to HIE systems reduces hospitalizations in EDs. Everson et al. (2017) demonstrate that the relationship between HIE and improved outcomes is mediated by faster accessing of information from outside organizations. Chen et al. (2019) show that participation in HIE networks reduces by 1.3% points the 30-day readmission rate for AMI patients. Ayer et al. (2019) find that HIE leads to a 10% reduction of patient length of stay (LOS) in EDs.

The relationship between HIE and service utilization has drawn much attention because HIE has been touted to result in reduced service utilization. However, empirical evidence is mixed. Lammers et al. (2014) find that HIE participation is associated with reduced repeat radiology imaging in the unaffiliated EDs within 30 days of previous visits. Similar results are shown by Nirel et al. (2010), Frisse et al. (2011), and Ross et al. (2013) under different settings. By contrast, there are other studies showing mixed effects of HIE on service utilization. For instance, McCormick et al. (2012) find that the electronic access to imaging results is associated with even higher imaging ordering rates. Eftekhari et al. (2017) show that a provider's tenure with HIE significantly lowers the repetition of therapeutic medical procedures, whereas no significant impact is found with diagnostic procedures. Ayabakan et al. (2017) demonstrate that HIE does not always lead to a reduction in duplicate testing. Moreover, the effects are varying among different types of tests—the reduction of duplicate testing is more pronounced for radiology tests than for laboratory tests.

Broadly speaking, our study is related to the literature on the use of information technology in healthcare delivery. With heavy spending by both governments and industries to promote the use of information technology in healthcare, it is expected that HIT holds the potential to have a profound effect on the way healthcare providers are operated and organized. According to Hydari et al. (2018), using advanced EHR systems leads to a decline in patient safety events up to 17.5%. Lu et al. (2018) find that an advanced EHR application, computerized physician order entry, in nursing homes enhances nursing task automation, resulting in better clinical quality and reduced staffing level in high-end facilities. Other than the aforementioned impacts, HIT adoption has been shown to reduce malpractice (Menon and Kohli 2013, Sharma et al. 2019), improve patient satisfaction (Queenan et al. 2011), and increase revenues (Devaraj et al. 2013). In addition to HIE and EHR, the impact of various technology innovations on care delivery, such as e-visits (Bavafa et al. 2018) and telemedicine (Delana et al. 2019, Sun et al. 2020), has gained attention.

Although the impact of various information technologies, including HIE, on healthcare delivery has been widely studied, almost all existing studies, including those mentioned earlier, examine it *within* healthcare organizations. To the best of our knowledge, there are only two studies investigating the factors affecting HIT interoperability *across* organizations. Li (2014) studies the factors that affect hospitals' decision on adopting interoperable EHR systems and shows that hospitals having a larger size of overlapping patient populations with interoperable hospitals are more likely to acquire interoperability. Using the network economics framework, Desai (2016) shows that although interoperability is valued, not all hospitals would choose to be interoperable with others, and to protect their markets, some hospitals choose information blocking to hinder interoperability. Clearly, our study distinguishes itself from the two HIT interoperability studies and the existing HIE literature by addressing the effect of interoperability on care delivery activities *across* different healthcare providers. Specifically, we study the throughput time and duplicate testing in interhospital patient transfers.

3.2. Patient Routing

To meet the challenge of heterogenous patient demand, effective and efficient patient routing through the healthcare system is critical to achieving a match between patients, healthcare providers, and resources (KC et al. 2020). As the primary decision maker of patients' access to healthcare resources, the role of hospitals is essential in routing patients. Hospitals' routing decisions, such as admission and discharge decisions, are greatly affected by available resources. For example, KC and Terwiesch (2017) find that limited inpatient bed availability leads to fewer inpatient admissions from EDs. High levels of resource occupancy lead to similar admission decisions in ICUs (Kim et al. 2015, Chan et al. 2017) and maternity units (Freeman et al. 2016). Meanwhile, when one resource fails to provide care in timely manner, routing patients is faced with resource escalation. The study by Chan et al. (2017) shows that delays in ICU admission subsequently lead to longer ICU length of stay. Such bounce-back effects are also found with health outcomes. For instance, shorter LOS and early discharges are associated with worse health outcomes (Oh et al. 2018).

In addition to admission and discharge policies, operations management scholars have investigated other policies in various healthcare settings, aiming to improve process efficiency and healthcare outcomes. For instance, Song et al. (2015) empirically investigate the benefits of dedicated queues in the context of ED patient management and show that they achieve shorter waiting time and LOS than pooled queues. KC and Terwiesch (2017) propose that scheduling patients uniformly over the week is an effective approach to enable more patients access to ED services. Hu et al. (2018) suggest that proactively transferring severe patients to the ICU generates lower mortality and shorter LOS. Chan et al. (2019) explore the role of step-down units (SDUs) to meet varying demand of patients. They find that SDUs are cost effective for post-ICU patients but not suitable for more severe patients, who need to be sent to ICUs.

Compared with routing patients within hospitals, coordinated healthcare delivery across multiple providers is more challenging in practice and is a research area less explored by researchers. Gupta et al. (2010) find inefficiency in interfacility transfers in the form of duplicate medical imaging, which indicates poor coordination between hospitals. Our research setting is similar to that of Lu and Lu (2018), who empirically investigate the factors that influence the choice of transfer destinations for heart attack patients. They find that compared with distance and quality, being affiliated with the same multihospital system plays a dominant role in determining transfer destinations.

Our paper contributes to the literature on patient routing by investigating the role of information technology in improving the efficiency of patient routing across different providers. Our findings provide a mechanism through which HIT interoperability improves patient outcomes. We also demonstrate that there exist barriers in realizing one of the promised benefits of HIT—reducing duplicate testing.

4. Data

Our study utilizes three data sources. First, we obtain the New York patient data from the State Emergency Department Database and the State Inpatient Database from 2013 to 2015. These data sets are available from the Healthcare Cost and Utility Project (HCUP) of the Agency for Healthcare Research and Quality. The data sets provide detailed patient information including age, health conditions, treatments, and time stamps for admissions at different facilities. Most importantly, they allow us to track a patient's historical visits to hospitals over time, thereby enabling us to identify those patients who were transferred between hospitals. Second, the Healthcare Information and Management Systems Society (HIMSS) database provides information about hospitals' adoption of various HIT software applications and their vendors. In 2013, HIMSS started to provide information about the adoption of HIE software applications. Third, we supplement the three main data sets with the American Hospital Association Annual Survey, which provides hospital characteristics including size, location, and affiliation (American Hospital Association 2018).

4.1. Measuring HIT Interoperability

Following the HITECH Act's specification of the meaningful use of HIT[10] and the recent literature on health information sharing (e.g., Ayer et al. 2019, Chen et al. 2019), we use the adoption of HIE to identify HIT interoperability. Namely, we define a pair of hospitals as interoperable if and only if both of them have live and operational HIE software applications, according to the HIMSS database. We denote this variable as *Interoperability*. The underlying rationale is that HIE software applications adopted by disparate healthcare providers enable them to exchange patient medical records using industry accepted standards.[11]

Generally, HIE software works by adding the interoperability functionality to existing EHR systems. The largest HIE software vendor in New York State, InterSystem, offers HIE software named HealthShare, which produces unified health records from different EHR systems to enable interoperability between healthcare providers. InterSystem's HealthShare has been deployed by several regions in New York State to enable community HIE networks. Meanwhile, EHR vendors have also been making efforts to embed the interoperability functionality in their own products. For example, Care Everywhere, the HIE software product of Epic, was introduced to integrate the interoperability functionality in Epic's own EHR system in 2009. In our sample, some hospitals in New York State have adopted Epic's EHR system and HIE software. This is an example of EHR-vendor-hub HIE.

Aside from this baseline definition of interoperability, which is a commonly used approach in the existing interoperability literature (e.g., Ayer et al. 2019), we will consider a second and more restrictive definition of interoperability by requiring a pair of hospitals to adopt an HIE software application offered by the *same* vendor. This will be explored as the heterogeneous analysis in Section 5.5. Our baseline definition of interoperability corresponds to any type of HIE software applications, regardless of their vendors. In our sample, hospitals have adopted HIE applications provided by community HIE vendors (e.g., Healthelink, Healthix, and InterSystem) and those provided by EHR software vendors (e.g., Cerner and Epic). Among all interhospital transfers of heart attack patients from 2013 to 2015, 52.32% of the patients were transferred between interoperable hospitals, and 22.6% of them were transferred between hospitals that adopted HIE applications from a common software vendor.

4.2. Identifying Interhospital Transfer of Heart Attack Patients

Using the HCUP data, we identify heart attack patients who were transferred from the ED of one hospital to another hospital. In the rest of the paper, we use "heart attack" and "AMI" interchangeably. AMI patients can be identified by the single Clinical Classifications Software code, which is equal to 100 under both the 9th version and 10th revision of the International Statistical Classification of Diseases and Related Health Problems (ICD). Patient identification is conducted based on the following rules.

• Same patient: patients with the same *VisitLink* in two records.

• Same day: discharge date in the sending hospital is the same as the admission date in the receiving hospital (*DaysToEvent* plus the LOS of the first record equal to the *DaysToEvent* of the second record).

• Being transferred: patients' disposition from the sending hospital is *being transferred to another hospital* (i.e., DISPUNIFORM = 2).

• Transfer between different hospitals: the sending and receiving hospitals have different identification numbers (i.e., DSHOSPIDs are different).

Details about how to identify transferred patients between hospitals are available in the HCUP User Guide: Supplemental Variables for Revisit Analysis.[12]

4.3. Measuring the Throughput Time of Interhospital Transfer

We define the throughput time of interhospital transfer to be the time interval between a patient's admission at the sending hospital's ED and his admission at the receiving hospital. For convenience, we will use "_x" to denote the time stamps in the sending hospital and "_y" to denote the time stamps in the receiving hospital, and the unit of measurement is hour. The throughput time of interhospital transfer is given by the following equation:

$$\textit{Throughput\_Time}\ (TPT) = \begin{cases} \text{AHOUR\_}y - \text{AHOUR\_}x, \text{ if LOS\_}x = 0; \\ 24 - \text{AHOUR\_}x + 24 \times (\text{LOS\_}x - 1) \\ \qquad + \text{AHOUR\_}y, \text{ if LOS\_}x > 0, \end{cases}$$

where AHOUR_x is the admission hour at the sending hospital, AHOUR_y is the admission hour at the receiving hospital, and LOS_x is the LOS at the sending hospital measured in days.

There exist some cases in our data set where the throughput time is within one hour. For these cases, the time intervals as calculated would be equal to zero because the time stamps are recorded in hours. To make our data reflect the reality better, we replace 0 hours with 0.5 hours. Our results are robust even without doing this correction. In addition, we drop cases with the throughput time longer than 24 hours, which account for 2% of all transfer cases. The estimated results using the full sample and samples generated using other outlier dropping rules are available in Tables A1 and A2 in the online appendix.

4.4. Identifying Duplicate EKG Testing

EKG utilization records in both sending and receiving hospitals are available in the New York State patient data. We identify a duplicate test if a patient receives a second EKG test in the receiving hospital despite that he has already received an EKG test in the sending hospital. Based on our sample, 86.8% of transferred AMI patients received duplicate EKG tests. Among those 4,960 transferred cases, 3,452 (69.6%) patients were diagnosed by the sending hospitals as STEMI and needed to undergo revascularization procedures immediately; 3,025 (87.6%) of the STEMI patients received duplicate EKG tests.

4.5. Measuring Health Outcomes

We adopt two health outcome measures in this study: 30-day all-cause readmission and in-hospital mortality. The 30-day all-cause readmission rate has been adopted by the Medicare Hospital Readmission Reduction Program (HRRP) for nationwide hospital quality improvement.[13] The HCUP data allow us to track patient history, and hence, we are able to identify those patients who returned to a hospital within 30 days. The in-hospital mortality rate is directly available from the HCUP data. Both variables are widely used to measure patient outcomes in the healthcare literature (e.g., Zhang et al. 2015, Lu and Rui 2018).

4.6. Control Variables and Sample Statistics

In our subsequent regression analysis, we control patient characteristics such as age, gender, race, insurance, location, and severity measured by the Charlson Comorbidity Index. For each pair of a sending hospital and a receiving hospital, we control their characteristics, including the indicator of whether they are affiliated with the same multihospital system, as well as their physical distance from each other. Furthermore, to control two hospitals' familiarity with each other in terms of handling AMI patient transfers, we include the number of AMI patients transferred between them in the previous year in all of our regressions.

Our main sample consists of 312 hospital pairs, which include 110 unique sending hospitals and 47 unique receiving hospitals. Most sending hospitals are community hospitals (106/110 = 96.4%), whereas most receiving hospitals are noncommunity hospitals (29/47 = 61.7%). Using the ICD procedure codes, we identify revascularization-capable hospitals as those that have conducted revascularization procedures (PCI/CABG) for at least 5% of the total heart disease patients (Cram et al. 2010, Lu and Lu 2018). In our sample of 110 sending hospitals, 96 do not have revascularization capability, among which most are community hospitals (92/96 = 95.8%). During our sample period of 2013–2015, 4,960 heart attack patients were transferred between hospitals in New York State. Among them, 3,452 were diagnosed as STEMI by their sending hospitals. We provide variable definitions and summary statistics in Table 1.

5. Impact of HIT Interoperability on Care Delivery Process Measures

5.1. Econometric Methods

The setting of interhospital transfer of heart attack patients provides a quasirandom assignment of patients to hospitals. Most heart attacks occur without any prior warnings, and local emergency services will be contacted to send patients to the nearest hospitals' EDs for rescue. As mandated by the U.S. Emergency Medical Treatment and Active Labor Act, hospitals cannot refuse to treat patients based on their insurance status. As a result, this process allows us to avoid patient selection of hospitals or hospital selection of patients.

5.1.1. The Throughput Time of Interhospital Transfer.

To estimate the effect of interoperability on the throughput time of interhospital transfer, we adopt the following specification:

$$\begin{aligned} \textit{Throughput\_Time}\ (TPT)_{ijpt} &= \alpha_0 + \alpha_1 \textit{Interoperability}_{ijt} \\ &\quad + \alpha_2 X_{pt} + \alpha_3 Z_{it} + \alpha_4 Z_{jt} \\ &\quad + \alpha_5 Z_{ijt} + \theta_i + \mu_t + \varepsilon_{ijpt}, \end{aligned} \tag{1}$$

Table 1. Summary Statistics

Variable	Definition	Number of observations	Mean	Standard deviation
Outcome measures				
throughput_time	Total time from being admitted in a sending hospital's ED to arriving at a receiving hospital (measured by hours)	4,960	5.3	3.92
dup_ekg	1 if a duplicate EKG test is done in a receiving hospital and 0 otherwise	4,960	0.87	0.34
died	1 if a patient dies in the hospital and 0 otherwise	4,960	0.05	0.23
thirty_day	1 if a patient is readmitted to a hospital within 30 days and 0 otherwise	4,960	0.20	0.40
Interoperability measure				
Interoperability	1 if the sending hospital and the receiving hospital are interoperable and 0 otherwise	4,960	0.52	0.50
Patient characteristics				
age	Patient's age	4,960	64.82	13.90
female	1 if a patient is female	4,960	0.37	0.48
white	Patient's race: White	4,960	0.84	0.37
black	Patient's race: Black	4,960	0.06	0.24
micropolitan	Patient's location: micropolitan	4,960	0.22	0.42
metroploitan	Patient's location: metropolitan	4,960	0.67	0.47
medicare	Expected primary payer type: Medicare	4,960	0.50	0.50
medicaid	Expected primary payer type: Medicaid	4,960	0.10	0.29
charlson index	Patient's comorbidity severity index	4,960	1.08	1.42
Hospital characteristics				
log_*nofbed_x*	Logarithm of the number of licensed beds in the sending hospital	4,960	4.95	0.83
log_*nofbed_y*	Logarithm of the number of licensed beds in the receiving hospital	4,960	6.12	0.42
academic_x	1 if the sending hospital is a teaching hospital and 0 otherwise	4,960	0.01	0.11
academic_y	1 if the receiving hospital is a teaching hospital and 0 otherwise	4,960	0.41	0.49
nonprofit_x	1 if the sending hospital is nonprofit and 0 otherwise	4,960	0.95	0.22
nonprofit_y	1 if the receiving hospital is nonprofit and 0 otherwise	4,960	0.86	0.35
ami_volume_month_x	Number of AMI patient admissions in the sending hospital ED	4,960	6.06	4.03
ami_volume_month_y	Number of AMI patient admissions in the receiving hospital	4,960	92.51	47.75
Hospital pair characteristics				
log_*dist*	Logarithm of the distance between the sending hospital and the receiving hospital	4,960	3.34	0.98
noof_transfer	Number of transferred AMI patients between hospital pair in the previous year	4,960	13.12	25.19
sys	1 if both hospitals are affiliated with the same hospital system	4,960	0.36	0.48
Other HIT applications				
hie_x	Indicator of HIE adoption in the sending hospital	4,960	0.62	0.49
telemedicine_x	Indicator of telemedicine adoption in the sending hospital	4,960	0.36	0.48
ehr_x	Indicator of EHR adoption in the sending hospital	4,960	0.96	0.18
edis_x	Indicator of EDIS adoption in the sending hospital	4,960	0.91	0.29

where $Throughput\_Time_{ijpt}$ represents the throughput time patient p spends in the interhospital transfer process, which starts from the time of admission in the sending hospital i's ED to the time of admission in the receiving hospital j at time t. $Interoperability_{ijt}$ is a binary variable measuring HIT interoperability, and it equals to one if hospital i and hospital j are interoperable at time t and equals zero otherwise.[14] In our baseline specification, we focus on any type of interoperability. We will explore interoperability enabled through a common HIE vendor as the heterogenous analysis in Section 5.5. The coefficient α_1 captures the effect of interoperability.

In the specification of Equation (1), we have carefully controlled for unobserved factors that might affect the transfer process. For example, θ_i is the fixed effect of the sending hospital, which controls for any unobserved factors of the sending hospital that might affect the transfer process. In addition, μ_t represents two time fixed effects, which include the year-quarter fixed effect and the weekend fixed effect. To further increase estimation precision, we include patient characteristics X_{pt} such as age, gender, race, insurance, and severity measured by the Charlson Comorbidity Index. We also incorporate time-varying controls at the hospital level: sending hospital characteristics Z_{it} and receiving hospital characteristics Z_{jt}, which include a hospital's academic status, nonprofit status, total number of beds, the monthly AMI patient admission volume of a sending hospital's ED, and the monthly AMI patient admission volume of receiving hospital (including both ED and inpatient). In addition to these hospital characteristics, we include a pairwise hospital control Z_{ijt}, which captures the physical distance between the sending hospital and the receiving hospital, whether they are affiliated with the same multihospital system, and the number of AMI patients transferred between hospital pairs in the previous year. ε_{ijpt} represents the error term. Standard errors are clustered by hospital pairs.

5.1.2. Duplicate EKG Testing.

We explore the effect of interoperability on duplicate EKG testing using the following specification:

$$dup\_ekg_{ijpt} = \alpha_0 + \alpha_1 Interoperability_{ijt} + \alpha_2 X_{pt} + \alpha_3 Z_{it} + \alpha_4 Z_{jt} + \alpha_5 Z_{ijt} + \theta_j + \mu_t + \varepsilon_{ijpt}, \quad (2)$$

where dup_ekg_{ijpt} is a binary variable that equals to one if patient p, who is transferred from hospital i to hospital j at time t, receives a duplicate EKG test in the receiving hospital j and equals zero otherwise. Because the decision of prescribing a duplicate EKG test takes place in the receiving hospital, we include fixed effect θ_j to capture any unobserved factors of the receiving hospital that may affect the decision of a duplicate EKG test. In this specification, μ_t represents two time fixed effects, which include the year-quarter fixed effect and the weekend fixed effect, whereas X_{pt} represents patient characteristics such as age, gender, race, insurance, and severity measured by the Charlson Comorbidity Index. In addition, we include sending hospital characteristics Z_{it} and receiving hospital characteristics Z_{jt} such as a hospital's academic status, nonprofit status, total number of beds, the monthly AMI patient admission volume of a sending hospital's ED, and the monthly AMI patient admission volume of the receiving hospital (including both ED and inpatient). We also include a pairwise hospital characteristic Z_{ijt}, which includes whether they are affiliated with the same multihospital system and the number of AMI patients transferred between hospital pairs in the previous year. ε_{ijpt} is the error term. Standard errors are clustered by hospital pairs.

5.2. The Effect of HIT Interoperability on the Throughput Time of Interhospital Transfer

We first examine the effect of interoperability on the throughput time of interhospital transfer. Table 2 shows the regression results for AMI patients using the throughput time or its log transformation as the dependent variable. Columns (1) and (2) show that the coefficients of interoperability are negative and significant, suggesting that compared with noninteroperable hospital pairs, the reduction of the throughput time is 0.760 hours (45.6 minutes) for AMI patients transferred between interoperable hospitals on average, and it accounts for 12.0% of the average throughput time. For STEMI patients (i.e., the more severe heart attack patients), the reduction of the throughput time is 0.646 hours (38.8 minutes) on average or 10.8% of the average throughput time (see Table 2, columns (4) and (5)). These results indicate that patients transferred between interoperable hospitals experienced a significantly shorter throughput time than patients transferred between noninteroperable hospitals, which supports Hypothesis 1.

5.3. The Effect of HIT Interoperability on Duplicate EKG Testing

We next explore the effect of interoperability on duplicate testing of transferred AMI patients. The results in Table 2 (columns (3) and (6)) show that duplicate EKG testing of AMI patients at receiving hospitals was not significantly reduced by interoperability, which does not support Hypothesis 2. As shown in column (6), even when a patient had been diagnosed by a sending hospital as STEMI and required revascularization procedures according to common cardiac care practices, interoperability did not significantly reduce a patient's chance of getting duplicate EKG testing at the

Table 2. Effects of HIT Interoperability: Main Results

	AMI			STEMI		
	(1) *TPT*	(2) log(*TPT*)	(3) *dup_ekg*	(4) *TPT*	(5) log(*TPT*)	(6) *dup_ekg*
Interoperability	−0.760**	−0.120***	−0.002	−0.646*	−0.108*	−0.005
	(0.230)	(0.034)	(0.015)	(0.253)	(0.042)	(0.018)
Patient characteristics	Y	Y	Y	Y	Y	Y
Hospital characteristics	Y	Y	Y	Y	Y	Y
Hospital pair characteristics	Y	Y	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y	Y	Y
Sending hospital FE	Y	Y	N	Y	Y	N
Receiving hospital FE	N	N	Y	N	N	Y
N	4,960	4,960	4,960	3,452	3,452	3,452
R^2	0.246	0.281	0.048	0.314	0.322	0.056

Notes. Standard errors in parentheses are clustered by hospital pairs. FE, fixed effect; N, no; TPT, throughput time; Y, yes.
$^+p < 0.1$; $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

receiving hospital. Given that the dependent variable, duplicate EKG testing, is a binary variable, we also ran a logistic regression model and obtained similar results. See Table A6 (columns (1)–(4)) in the online appendix.

Although reducing duplicate tests and procedures is touted as one of the key benefits of EHR, our finding suggests that in practice, this promised benefit is not easily attained even when hospitals have acquired interoperability. We provide several explanations for the results. First, any testing results are associated with type I and type II errors. To reduce the risk of misdiagnosis, it is reasonable for a receiving hospital to repeat tests. For example, it has been reported that 11.0% of STEMI patients had inconclusive initial EKG testing, among whom 72.4% were later diagnosed as STEMI by a follow-up EKG within 90 minutes of their initial EKG (Riley et al. 2013).

Second, the conditions of heart attack patients are not static but dynamic and evolving (Feazel et al. 2015). If the initial EKG was inconclusive, the guideline of the American College of Cardiology/American Heart Association suggests obtaining either serial EKGs at 5- to 10-minute intervals or continuous 12-lead ST-segment monitoring in order to detect the development of ST elevation.[15] Therefore, repeat testing might be necessary.

The last possibility stems from the gap between HIE adoption and HIE utilization in practice. According to Johnson et al. (2011), on average the utilization of HIE in ED is 6.8%. Some evidence shows that even if required information is returned via HIE, not all information is viewed by clinicians (Everson et al. 2017). Based on a nationally representative sample, McCormick et al. (2012) find that computerized access to imaging and laboratory results is surprisingly associated with more ordering of imaging and blood tests, which could be because of low utilization of prior test results.

Given no significant reduction in duplicate EKG testing, one may wonder whether it is driven by the fact that EKG is a relatively quick, painless, and inexpensive test. To address this issue, we examine the effect of HIT interoperability on duplicate testing of CT scan, which was used on 13.0% of all transferred heart attack patients in sending hospitals, according to our data. CT scan not only takes a longer time than EKG but also, exposes patients to low-dose radiation. Moreover, the cost of heart CT scan ranges from $1,400 to $3,600, which is much higher than the average cost of EKG ($595) in New York State.[16] We report our results in Table A7 in the online appendix, which shows that HIT interoperability still has no statistically significant effect on duplicate testing of CT scan.

5.4. Robustness Check

In this section, we conduct various robustness checks to validate the main results reported in Sections 6.1 and 6.2. In short, the results of these robustness checks are consistent with our main results.

5.4.1. Alternative Specifications. First, we estimate Equations (1) and (2) with both the sending hospital fixed effect and the receiving hospital fixed effect as a robustness check because interhospital transfers involve both sending and receiving hospitals. The results in Table 3 indicate that after including these two types of hospital fixed effects, the estimated coefficients are similar to our main results. Specifically, we find that HIT interoperability leads to an average throughput time reductions of 36.8 (9.8%) and 32.9 minutes (9.4%) for AMI patients and STEMI patients, respectively, but has no significant effect on duplicate EKG testing.

Second, to rule out the concern that our results might be driven by time-variant unobserved differences between interoperable and noninteroperable sending hospitals, we construct a subsample of hospital pairs in which all sending hospitals transferred patients to both interoperable and noninteroperable

Table 3. Effects of HIT Interoperability: Two Fixed Effects

	AMI			STEMI		
	(1) *TPT*	(2) log(*TPT*)	(3) *dup_ekg*	(4) *TPT*	(5) log(*TPT*)	(6) *dup_ekg*
Interoperability	−0.614**	−0.098**	0.003	−0.549*	−0.094*	−0.005
	(0.224)	(0.033)	(0.022)	(0.223)	(0.038)	(0.027)
Patient characteristics	Y	Y	Y	Y	Y	Y
Hospital characteristics	Y	Y	Y	Y	Y	Y
Hospital pair characteristics	Y	Y	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y	Y	Y
Sending hospital FE	Y	Y	Y	Y	Y	Y
Receiving hospital FE	Y	Y	Y	Y	Y	Y
N	4,960	4,960	4,960	3,452	3,452	3,452
R^2	0.278	0.307	0.104	0.344	0.349	0.125

Notes. Standard errors in parentheses are clustered by hospital pairs. FE, fixed effect; N, no; TPT, throughput time; Y, yes.
$^+p < 0.1$; $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

receiving hospitals (i.e., to explore the within variation of sending hospitals). With this subsample, we conduct the regressions and report the results in Table A3 (columns (1)–(3)) in the online appendix. The results are qualitatively and quantitatively similar to our main results.

Third, as we stated in Section 2.1, because of lack of availability of revascularization capability, heart attack patients are often transferred to revascularization-capable hospitals to get more advanced treatments. Therefore, we construct a subsample of patients transferred from nonrevascularization hospitals to revascularization hospitals, and they account for 77.6% of all transferred patients. The estimated results are shown in Table A3 (columns (4)–(6)) in the online appendix. They indicate that HIT interoperability leads to 0.97 hours (13.5%) reduction in the throughput time but exerts no significant effect on duplicate EKG testing, which is consistent with our main results.

5.4.2. Effects of Other HIT Applications on Interhospital Transfer. Existing studies have shown that the adoption of various HIT applications, such as telemedicine (Sun et al. 2020), EHR (Oh et al. 2018), and HIE (Ayer et al. 2019) in ED, would lead to efficiency improvement in healthcare delivery and reduction of patients' length of stay. To isolate the effect of HIT interoperability, we need to consider the effect of other HIT applications' adoption in sending hospitals. Thus, we add extra controls for HIT adoption in sending hospitals to our regression model such as telemedicine, EHR, emergency department information system (EDIS), and HIE. One caveat is that we cannot include all these control variables together in one regression because of multicollinearity between them. Therefore, we conduct regressions by controlling one type of HIT application at a time and report the results in Table 4.

The results in Table 4 show that after controlling the effects of other HIT applications' adoption in sending hospitals, the results remain consistent with our main results—the magnitude of the reduction in the throughput time ranges from 9.9% to 12.3%. In contrast, the coefficients of the other HIT applications are not statistically significant, which implies that their effects on the throughput time are insignificant.

5.4.3. Instrumental Variable for Interoperability. In this study, we assume that heart attack patients are quasirandomly assigned to different hospitals with or without interoperability. This assumption is reasonable given the fact that emergency ambulance service would take patients to the nearest ED, mitigating the hospital selection issue. Despite this, we have included the hospital fixed effects in our regression model to control for any time-invariant unobserved factors that might affect both the adoption of interoperability and the transfer process. Nevertheless, one may be concerned with the existence of unobserved time-varying factors causing endogeneity issues. Therefore, we will use the IV approach to further address this issue.

To proceed, we propose an IV approach inspired by the network effect of HIT (Miller and Tucker 2009). It is believed that as more neighboring hospitals adopt HIT, it is more likely for a hospital to adopt HIT because of benefits such as learning effects (Karshenas and Stoneman 1993) and cost sharing (Dranove et al. 2014). Similarly, we believe that interoperability has such a network effect, which allows us to develop an IV for interoperability. To do so, we calculate the average adoption rate of interoperability in the same health service area (HSA) region[17] for both sending hospitals and receiving hospitals separately. The calculation *excludes* sending and receiving hospitals themselves. Because interoperability is defined between a pair of hospitals, we use the product of the average adoption rates of the two hospitals as the IV

Table 4. Effects of Other HIT Applications on Throughput Time

	(1) *TPT*	(2) log(*TPT*)	(3) *TPT*	(4) log(*TPT*)	(5) *TPT*	(6) log(*TPT*)	(7) *TPT*	(8) log(*TPT*)
Interoperability	−0.805***	−0.123***	−0.787**	−0.122***	−0.726**	−0.114**	−0.608⁺	−0.099⁺
	(0.239)	(0.035)	(0.238)	(0.036)	(0.243)	(0.037)	(0.317)	(0.053)
Telemedicine_*x*	0.230	0.015						
	(0.273)	(0.053)						
EHR_*x*			0.370	0.032				
			(0.593)	(0.085)				
EDIS_*x*					−0.357	−0.065		
					(0.354)	(0.068)		
HIE_*x*							−0.256	−0.035
							(0.366)	(0.062)
Patient characteristics	Y	Y	Y	Y	Y	Y	Y	Y
Hospital characteristics	Y	Y	Y	Y	Y	Y	Y	Y
Hospital pair characteristics	Y	Y	Y	Y	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y	Y	Y	Y	Y
Sending hospital FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	4,960	4,960	4,960	4,960	4,960	4,960	4,960	4,960
R^2	0.246	0.281	0.246	0.281	0.246	0.281	0.246	0.281

Notes. Standard errors in parentheses are clustered by hospital pairs. FE, fixed effect; N, no; TPT, throughput time; Y, yes.
$^+p < 0.1$; $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

for interoperability:

$$IV_{ijt} = \frac{\begin{array}{c}\textit{Number of adopters in the neighboring}\\ \textit{region of sending hospital } i \textit{ in year } t\end{array}}{\begin{array}{c}(\textit{Total number of hospitals in the neighboring}\\ \textit{region of sending hospital } i) - 1\end{array}} \times \frac{\begin{array}{c}\textit{Number of adopters in the neighboring}\\ \textit{region of receiving hospital } j \textit{ in year } t\end{array}}{\begin{array}{c}(\textit{Total number of hospitals in the neighboring}\\ \textit{region of receiving hospital } j) - 1\end{array}}.$$

The underlying assumption of this IV is that the rate of interoperability adoption of neighboring hospitals is correlated with the adoption of interoperability of sending and receiving hospitals but does not directly affect the healthcare delivery process for patients in these hospitals. This IV approach was adopted by Lu et al. (2018).

Using this IV approach, we conduct two-stage least square (2SLS) regressions. Columns (1) and (2) in Table 5 show that interoperability leads to a reduction of 1.072 hours (64.3 minutes) in the throughput time or 17.2%. These 2SLS results are qualitatively and quantitatively consistent with the results in Table 2. In addition, column (3) in Table 5 indicates that interoperability has no significant effect on duplicate EKG testing, which is also consistent with the result in Table 2.

5.4.4. Propensity Score Matching. As described in Section 2.3, the decision on whether to transfer heart attack patients is at the discretion of doctors, who can choose to administer clot-busting medication on the patients and monitor their conditions at the original hospitals. To mitigate any potential bias caused by doctors' preference, we conduct patient-level PSM to validate our results. The PSM approach is widely used in many literatures such as healthcare, marketing, and operations to mitigate potential selection bias.

To perform patient-level propensity score matching, we define patients who are sent between interoperable hospitals as the treatment group. The primary purpose of matching is to find the control group of patients who share similar characteristics as the treatment patients but are not transferred between interoperable hospitals. We use a logit model to compute propensity scores, and the explanatory variables are patient characteristics, such as demographic and illness severity, and hospital characteristics, such as number of beds.

Using propensity score matching with nearest neighbors specified as three and a caliper of 0.25 standard deviation of the propensity score, we get a sample with size 4,719. The performance of matching is reported as s balance check in Table A4 in the online appendix, which compares the characteristics of the treatment and the control groups. There are no statistically significant differences in all characteristics between the two groups. Using this matched sample, we reestimate the models as specified by Equations (1) and (2). The results are displayed in Table 5, columns (4)–(6), which shows that interoperability leads to a reduction of 0.724 hours (43.4 minutes) in the throughput time or 11.7% but has no significant effect on duplicate EKG testing. These results are similar to those in Table 2, confirming the robustness of our main results.

Table 5. Effects of HIT Interoperability: Robustness Check

	2SLS			PSM			DID		
	(1) *TPT*	(2) log(*TPT*)	(3) *dup_ekg*	(4) *TPT*	(5) log(*TPT*)	(6) *dup_ekg*	(7) *TPT*	(8) log(*TPT*)	(9) *dup_ekg*
Interoperability	−1.072***	−0.172***	−0.005	−0.724**	−0.117***	−0.005	−0.811**	−0.118**	−0.007
	(0.353)	(0.055)	(0.024)	(0.229)	(0.034)	(0.016)	(0.277)	(0.041)	(0.024)
First-stage IV	0.791***	0.791***	0.868***						
	(0.087)	(0.087)	(0.069)						
Patient characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hospital characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hospital pair characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sending hospital FE	Y	Y	N	Y	Y	N	Y	Y	N
Receiving hospital FE	N	N	Y	N	N	Y	N	N	Y
First-stage *F* statistic	3,430.338	3,430.338	6,175.222						
N	4,798	4,798	4,798	4,719	4,719	4,719	3,910	3,910	3,910
R^2	0.250	0.282	0.049	0.242	0.279	0.047	0.265	0.297	0.086

Notes. Standard errors in parentheses are clustered by hospital pairs. FE, fixed effect; N, no; TPT, throughput time; Y, yes.
$^{+}p < 0.1$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

5.4.5. Difference-in-Differences Analysis. In this section, we conduct a DID analysis by exploring the within-hospital variation in the adoption of HIT interoperability. This analysis serves as a robustness check for our main results. It is noteworthy that HIMSS only started to provide hospital HIE adoption information in 2013. Therefore, we cannot obtain a data set with a longer time span to address the parallel pretrend issue. In our data, which are dated from 2013 to 2015, there are three types of sending and receiving hospital pairs: (1) always interoperable hospital pairs (i.e., those that became interoperable in 2013 or before), (2) newly interoperable hospital pairs (i.e., those that became interoperable in 2014 or 2015), and (3) noninteroperable hospital pairs (i.e., those that had not become interoperable by the end of 2015). To conduct a DID analysis, we construct a subsample that only includes newly interoperable hospital pairs and noninteroperable hospital pairs. With this subsample, the two-way fixed effect models specified by Equations (1) and (2) would yield standard DID results because variations in the hospital adoption status help to identify the effect of HIT interoperability. The results are displayed in Table 5, columns (7)–(9), which shows that HIT interoperability leads to a reduction of 0.811 hours (48.7 minutes) in the throughput time of interhospital transfer or 11.8%. Also, no significant effect is found on duplicate EKG testing. These results are consistent with the main results presented in Table 2.

5.4.6. Falsification Test: ED Efficiency Improvement. We conduct a falsification test to rule out the possibility that the reduction in the throughput time of interhospital transfer is because of ED efficiency improvement that is *not* caused by the adoption of interoperability. The Hospital Compare data set provides two hospital ED efficiency performance indices, ED-1b and ED-2b. ED-1b is defined as the average time patients spend in the ED before they are admitted to the hospital as an inpatient, whereas ED-2b is defined as the average time patients spend in the ED after the attending doctor has decided to admit them as inpatients but before leaving the ED for their inpatient wards. To carry out the falsification test, we construct a sample that only includes patients who are transferred between *noninteroperable* hospital pairs. We assign all sending hospitals with the two ED efficiency performance indices *lower* than the median as "interoperable" (because higher ED-1b or ED-2b index value means lower ED efficiency). If ED efficiency in sending hospitals does play a significant role in speeding up the interhospital transfer process, we shall expect to see that efficient EDs are associated with a significant reduction in the throughput time. The results are reported in Table A5 in the online appendix, which shows that the hypothetically assigned interoperability (based on ED efficiency measures ED-1b and ED-2b) does not have a statistically significant effect on the throughput time and duplicate EKG testing. This falsification test demonstrates that it is interoperability, not ED efficiency, that drives the reduction in the throughput time of interhospital transfer.

5.5. Heterogeneous Effect: Common HIE Vendor

Although in theory, interoperable providers can exchange patient records with any other interoperable providers, having HIE applications from a common vendor is believed to have a higher level of efficiency for interfacility activities because of the naturally similar technology infrastructure and higher compatibility, according to some existing studies (Li 2014, Desai 2016). For instance, when a clinician queries information via Care Everywhere, an HIE application created by the biggest HIT vendor in the United States (Epic) to request patient information from an outside organization, the clinician can view the information using her hospital's EHR if the outside organization is also connected to Care Everywhere. The returned information is structured and comes with patient record links to all available patient information that goes beyond the requested information. In contrast, if the outside organization is not connected to Care Everywhere, the requested information will be scanned and returned as a PDF document, which provides limited information and cannot be easily digitalized and saved into the requester's EHR system (Everson et al. 2017). Inspired by these observations, in this section, we explore the heterogeneous effect of different degrees of interoperability. Specifically, we use HIE software vendor information in the HIMSS data set to identify whether two interoperable hospitals have adopted their HIE applications from the same vendor. Specifically, we define a binary variable, *Same_HIE_Vendor*, which equals one if two hospitals use an HIE software application provided by the same vendor and zero otherwise.[18]

To conduct the heterogeneous analysis, we add the control variable *Interoperability*× *Same_HIE_Vendor* in the regression models specified by Equations (1) and (2) and report the results in Table 6. As the table shows, interoperability enabled through a common HIE vendor yields 15.6% (or 49.9 minutes) more reduction in the throughput time than interoperability enabled through different HIE vendors. Meanwhile, no significant effect is found on duplicate EKG testing even when interoperability is enabled through a common vendor.

Table 6. Heterogeneous Effect: Common HIE Vendor

	(1) *TPT*	(2) log(*TPT*)	(3) *dup_ekg*
Interoperability	−0.569*	−0.084*	−0.002
	(0.250)	(0.037)	(0.017)
Interoperability × *Same_HIE_Vendor*	−0.832*	−0.156*	−0.001
	(0.368)	(0.063)	(0.029)
Patient characteristics	Y	Y	Y
Hospital characteristics	Y	Y	Y
Hospital pair characteristics	Y	Y	Y
Time FEs	Y	Y	Y
Sending hospital FE	Y	Y	N
Receiving hospital FE	N	N	Y
N	4,960	4,960	4,960
R^2	0.248	0.283	0.048

Notes. Standard errors in parentheses are clustered by hospital pairs. FE, fixed effect; N, no; TPT, throughput time; Y, yes.
$^+p < 0.1$; $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

To sum up, we have shown that when hospitals obtain interoperability by adopting HIE applications, they may experience different levels of efficiency improvement in their care delivery processes—when their HIE applications are provided by the same vendor, a more significant efficiency gain can be achieved in interhospital patient transfers.

6. Impact of HIT Interoperability on Health Outcomes

6.1. Effects of HIT Interoperability on 30-Day Readmission and In-Hospital Mortality

So far, we have shown that HIT interoperability leads to an efficiency improvement in the care for heart attack patients by speeding up the interhospital transfer process. Our results naturally give rise to the question of whether interoperability would improve health outcomes. An ideal way to answer the question is to conduct an experiment in which we assign patients with different health conditions to interoperable hospitals and noninteroperable ones randomly. Conducting such an experiment is not feasible. However, the heart attack setting provides a quasirandom assignment of patients to hospitals. Leveraging this quasiexperiment setting, we investigate the relationship between interoperability and health outcomes using the following specification:

$$Y_{ijpt} = \alpha_0 + \alpha_1 Interoperability_{ijt} + \alpha_2 X_{pt} + \alpha_3 Z_{it} + \alpha_4 Z_{jt} + \alpha_5 Z_{ijt} + \theta_i + \theta_j + \mu_t + \varepsilon_{ijpt}. \tag{3}$$

In Equation (3), Y_{ijpt} indicates the health outcome of patient p transferred from hospital i to hospital j at time t. We use two health outcome measures as the dependent variable, 30-day all-cause readmission denoted by $thirty\_day_{ijpt}$ and in-hospital mortality denoted by $died_{ijpt}$. $Thirty\_day_{ijpt}$ is a binary variable that equals one if patient p is readmitted to any hospital within 30 days after being discharged from hospital j and equals zero otherwise. $Died_{ijpt}$ is a binary variable that equals one if patient p dies in hospital j at time t and equals zero otherwise. Because a transferred patient gets treatment in both the sending and the receiving hospitals, both hospitals might affect the healthcare outcome of the patient. Hence, Equation (3) incorporates the fixed effects of sending hospitals (θ_i) and receiving hospitals (θ_j). Equation (3) also includes two time fixed effects (μ_t), which are the year-quarter fixed effect and the weekend fixed effect. For the time-variant controls, we include patient characteristics X_{pt} (*age*, *female*, *race*, *location*, *payer*, *charlson_index*), sending hospital characteristics Z_{it} (log_*nobed_x*, *academic_x*, *nonprofit_x*, *ami_volume_month_x*), receiving hospital characteristics Z_{jt} (log_*nobed_y*, *academic_y*, *nonprofit_y*, *ami_volume_month_y*), and hospital pair characteristics Z_{ijt} (*sys*, *noof_transfer*). ε_{ijpt} is the error term, and standard errors are clustered by receiving hospitals.

To demonstrate the robustness of our results, we apply the IV approach as described in Section 5.4.3, the PSM approach as described in Section 5.4.4, and the DID approach as described in Section 5.4.5. We also used a subsample consisting of patients transferred from nonrevascularization hospitals to revascularization hospitals, as described in Section 5.4.1. The regression results are reported in Table 7. Column (1) shows the baseline OLS result, and it indicates that interoperability leads to a 3.0-percentage point decrease in the all-cause 30-day readmission rate. Using the IV approach, we find the decrease is 7.8% points. The rest of the robustness checks yields a decrease ranging from 2.5 to 5.3% points. However, we observe no statistically significant change in the in-hospital mortality rate under all specifications. Taken together, our results provide some evidence to support Hypothesis 3.

6.2. Mechanism Discussion: Throughput Time and Health Outcomes

Now, the question is why interoperability would lead to an improved 30-day readmission rate for transferred patients. We provide an explanation from the lens of throughput time reduction. We have shown that interoperability reduces the throughput time of interhospital transfer. After a patient experiences heart attack, time is muscle—as every minute goes by, more heart muscle is going to die. It is vital to remove the blockage in the patient's heart arteries as soon as possible. Therefore, shortening the throughput time would improve the health outcome for heart attack patients, which has been demonstrated in existing medical studies (Chan et al. 2012, Estévez-Loureiro et al. 2013, Farshid et al. 2015).

We will test this claim using our sample to estimate the effect of throughput time on 30-day readmission rates. There exists an endogeneity issue in this test, which we need to carefully address. For example, some patient symptoms were observed by doctors but were not recorded in the New York patient data set; thus, they were not observable by researchers. These unobserved patient symptoms not only affect the throughput time by influencing doctors' decisions on when to transfer patients but also, affect patient outcomes. Therefore, we adopt an IV approach to mitigate this endogeneity issue.

To proceed, we calculate transferred patients' average duration of stay in the sending hospital for each sending and receiving hospital pair in the previous year and construct the variable *average_duration*. Then, we use *average_duration* and the logarithm of distance between the sending and the receiving hospitals (log_*dist*) as IVs for the throughput time of interhospital transfer. Because both the duration in the sending hospital and the transportation time are included in the throughput time, the two IVs are highly correlated with the throughput time, but they do not directly affect patient outcomes. With these IVs, we apply the 2SLS method on 30-day readmission rates and report the results in Table A8 in the online appendix. It shows that on average a one-hour reduction in the throughput time would lead to an 8.1-percentage point decrease in the 30-day readmission rates for transferred AMI patients. This result seems to suggest that throughput time reduction explains almost entirely the decrease of the 30-day readmission rate

Table 7. Effects of HIT Interoperability on Health Outcomes

	thirty_day					*died*				
	OLS (1)	2SLS (2)	PSM (3)	DID (4)	Subsample (6)	OLS (7)	2SLS (9)	PSM (10)	DID (11)	Subsample (12)
Interoperability	−0.030^{+}	−0.078***	−0.031^{+}	−0.025^{+}	−0.053**	−0.007	−0.005	−0.007	−0.008	−0.016
	(0.016)	(0.024)	(0.017)	(0.014)	(0.019)	(0.008)	(0.011)	(0.009)	(0.009)	(0.013)
First-stage IV		0.832***					0.832***			
		(0.089)					(0.089)			
Patient characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hospital characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Hospital pair characteristics	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sending hospital FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Receiving hospital FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
First-stage *F* statistic		3,968.289					3,968.289			
Observations	4,960	4,798	4,719	3,910	3,849	4,960	4,798	4,719	3,910	3,849
R^2	0.054	0.045	0.056	0.060	0.058	0.061	0.062	0.066	0.066	0.070

Notes. Subsample refers to the sample of patients sent from nonrevascularization hospitals to revascularization hospitals. Standard errors in parentheses are clustered by receiving hospitals. FE, fixed effect; N, no; Y, yes.
$^{+}p < 0.1$; $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

because of HIT interoperability (recall that Table 2 shows that interoperability leads to a reduction of 0.76 hours in the throughput time and that Table 7 shows that interoperability leads to a 2.5- to 7.8-percentage point decrease in the 30-day readmission rate). With this, we have established evidence to support the conjecture that throughput time reduction would lead to improved outcomes for transferred patients.

7. Conclusion and Implications

In this study, we empirically investigate the impact of HIT interoperability on interhospital transfer of heart attack patients. We find that HIT interoperability leads to an efficiency improvement in the interhospital transfer process by shortening by 45.6 minutes or 12.0% the throughput time on average but exerts no significant effect on duplicate EKG testing. In addition, when interoperability is enabled through a common HIE vendor, it yields 15.6% more reduction in the throughput time than when it is enabled through different vendors. Furthermore, we find that HIT interoperability leads to a 3.0-percentage point decrease in the 30-day all-cause readmission rate of transferred patients, suggesting a significant positive impact of HIT interoperability on patient outcomes.

Our study aims to investigate the impact of HIT interoperability on care delivery processes across providers through the lenses of patient transfer throughput time and duplicate testing. The evidence gathered from our study highlights the operational drivers through which HIT interoperability, a technological advancement in the health information system, could lead to better patient outcomes. Compared with existing studies on the effect of HIT *within hospitals*, we investigate the effect of HIT *across hospitals*. Understanding processes beyond the boundary of healthcare organizations is crucial to uncovering mechanisms for meaningful use of HIT. To the best of our knowledge, empirical studies on crosshospital processes are almost nonexistent in the existing HIT literature.

Our study reveals important actionable insights relevant to the practice. For hospitals, our findings suggest that HIT interoperability improves the efficiency of care coordination between hospitals by reducing heart attack patients' throughput time of interhospital transfer. For hospitals struggling with slow interhospital transfers, participation in HIE networks and enhancing their HIT interoperability with neighboring hospitals could be a way of improving the efficiency of care delivery. Under the mandate of the HRRP, hospitals will be financially penalized when their 30-day readmission rates go above the national average. Our study suggests that adopting HIE may help to reduce a hospital's AMI 30-day readmission rate, which is a condition-specific measure tracked by HRRP.[19]

We conduct a back-of-the-envelope analysis on the implications of HIT interoperability on healthcare expenditure. According to Khera et al. (2017), the annual AMI patient hospitalization was approximately 0.52 million in the United States in 2013, and the average cost of readmissions within 30 days after an AMI episode was estimated to be $14,885. We assume that the percentage of transfers among all heart attack patients admitted into EDs is 12.4% (Kindermann et al. 2013). Hence, the annual national volume of transferred AMI patients is roughly 64,000 (0.52 million × 12.4%). Based on our OLS and 2SLS estimations, HIT interoperability leads to a 3.0- or 7.8-percentage point decrease in the 30-day readmission rate. Therefore, we estimate that the magnitude of the readmission cost savings stemming from HIT interoperability is approximately $28.8–$74.9 million annually (64,000 × $14,885 × 3.0% or 7.8%) in the United States

For policy makers, our findings demonstrate the value of incentivizing HIT adoption and promoting widespread exchange of health information because HIT interoperability indeed improves the efficiency of healthcare delivery across providers, which ultimately translates to improved patient outcomes. Although the U.S. Government has promoted the adoption and meaningful use of HIT for over a decade, there is still a long way to achieve widespread HIT interoperability nationwide and make full use of it, as demonstrated by the lack of reduction in duplicate testing in our study. We call for more provider effort toward realizing the complete potential of HIT interoperability by minimizing the gap between technology adoption and utilization.

Our research is not without limitations. Our data set does not provide detailed time stamps for all activities during the interhospital transfer process, thereby preventing us from evaluating the impact of HIT interoperability on specific activities during crosshospital care delivery processes (e.g., time spent on heart attack diagnosis and time spent on the negotiation between hospitals). If more detailed process data are available, a finer analysis can be conducted on care delivery activities within and across hospitals. Nevertheless, we hope our study can spark an interest among healthcare operations researchers to explore this promising area of research in the future.

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Endnotes

[1] See https://www.healthit.gov/topic/health-it-and-health-information-exchange-basics/benefits-ehrs.

[2] See https://www.cms.gov/Regulations-and-Guidance/Legislation/EHRIncentivePrograms/index?redirect=/EHRIncentivePrograms.

[3] This was the Medicare Access and CHIP Reauthorization Act of 2015 (Public Law 114-10 Sec 106).

[4] See https://www.healthit.gov/sites/default/files/hie-interoperability/nationwide-interoperability-roadmap-final-version-1.0.pdf.

[5] This is because of the mandate of the U.S. Emergency Medical Treatment and Labor Act.

[6] See https://www.heart.org/en/health-topics/heart-attack/about-heart-attacks.

[7] Although SHIN-NY has "statewide" in its name, it did not achieve a statewide patient record query system until after 2015.

[8] In general, if a patient is diagnosed as STEMI from EKG testing, the patient will be transferred to a revascularization-capable hospital. If, however, the patient is not diagnosed as STEMI, the transfer decision will be at the discretion of the ED physician. When not transferred, a patient will be administered clot-busting drugs to alleviate the symptom.

[9] See https://www.miemss.org/home/Portals/0/Docs/Guidelines_Protocols/Interhospital-Transfer-Manual-20210323.pdf?ver=2021-03-24-083008-247.

[10] See https://www.cdc.gov/ehrmeaningfuluse/introduction.html.

[11] See https://www.champsoftware.com/2018/02/12/a-basic-overview-of-health-information-exchange/.

[12] See https://www.hcup-us.ahrq.gov/toolssoftware/revisit/revisit.jsp.

[13] See http://kff.org/medicare/issue-brief/aiming-for-fewer-hospital-u-turns-the-medicare-hospital-readmission-reduction-program/.

[14] Note that our data set only has year-level observations for all hospital characteristics, including interoperability.

[15] See https://www.ahajournals.org/doi/pdf/10.1161/circ.110.9.e82.

[16] See https://www.newchoicehealth.com/places/new-york/new-york/.

[17] The Dartmouth Atlas Project defines an HSA as "a collection of zip codes whose residents receive most of their hospitalizations from the hospitals in that area" (https://www.dartmouthatlas.org).

[18] If hospitals have missing HIE vendor information, we treat *Same_HIE_Vendor* = 0.

[19] See https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/AcuteInpatientPPS/Readmissions-Reduction-Program.

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